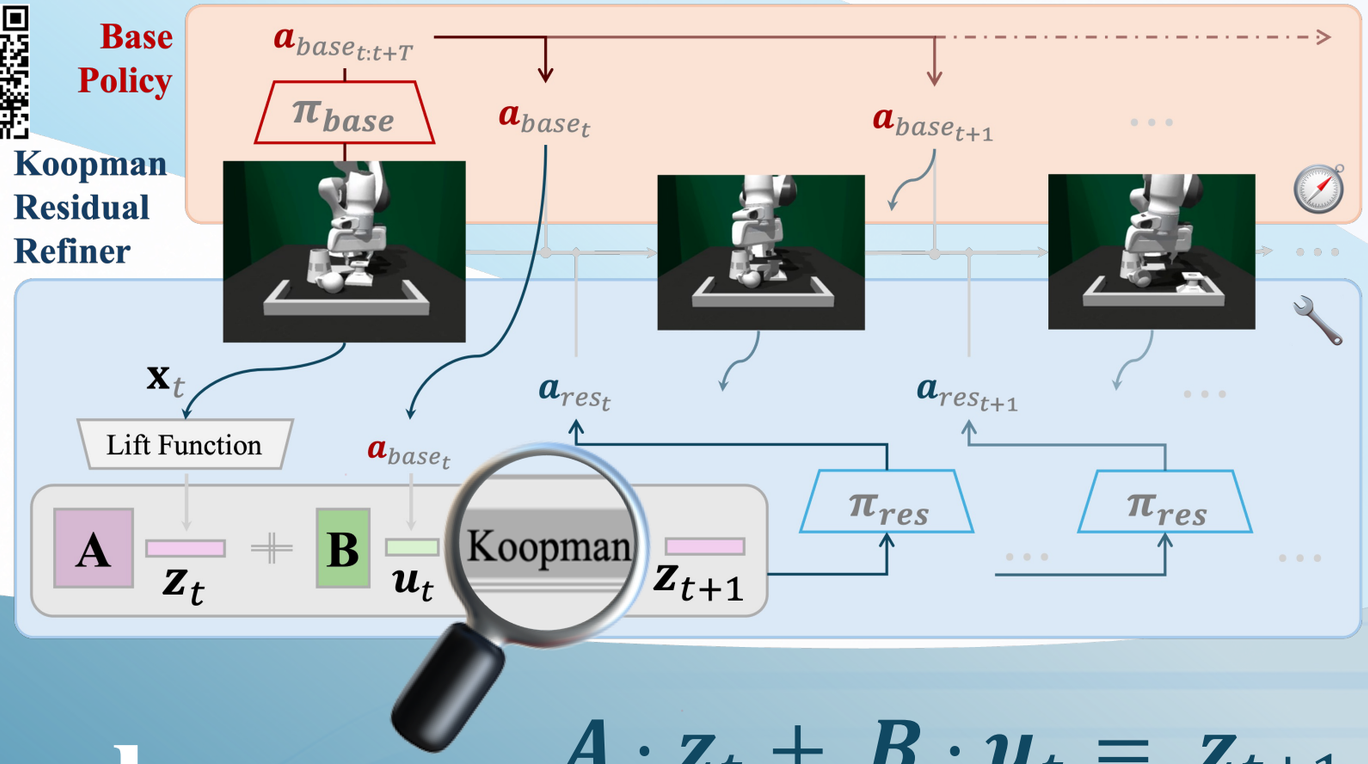
ICRA 2026 VIENNA

IEEE INTERNATIONAL CONFERENCE ON ROBOTICS AND AUTOMATION

Robust Online Residual Refinement via Koopman-Guided Dynamics Modeling

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Koopman-guided Online Residual Refinement

$$A \cdot z_t + B \cdot u_t = z_{t+1}$$

Model-Agnostic

Robust & General

Method

The **base policy** predicts a chunk of base actions at a lower frequency, while **KORR** refines these actions step-by-step at a higher control rate. For each base action a_{base_t}, **KORR** extrapolates the next possible state z_{t+1} using **Koopman** dynamics as an imagined future state, and then conditions the **residual policy** on this state to generate a corrective residual action a_{res_t} . The final executed action, obtained by combining the base and residual actions, forms a closed-loop refinement that enhances **robustness** and **generalization**.

Abstract

Imitation learning struggles with long-horizon and high-precision manipulation due to compounding errors. Existing residual refinement methods mainly rely on local corrections to the base policy output, limiting robustness and generalization.
We propose a **Koopman**-guided framework that leverages linear latent dynamics for globally informed and stable action correction within RL-based refinement.
Our findings further highlight the potential of Koopman-based modeling to bridge modern learning methods with classical control theory.

Experiment

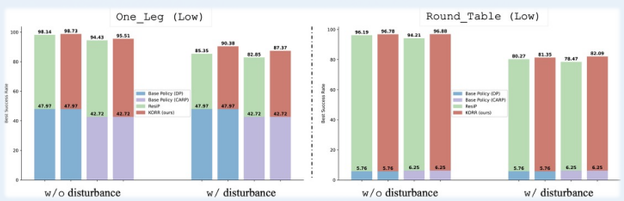
General Robustness and Performance Study

Methods	One_Leg						Round_Table				Lamp			
	Low		Med		High		Low		Med		Low		Med	
DP [2]	47.97		23.93		4.39		5.76		1.56		5.57		1.46	
	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/
ResiP [32]	98.14	85.35	84.99	46.09	30.03	3.20	96.19	80.27	52.73	22.66	86.58	60.55	51.66	29.98
KORR (ours)	98.73	**90.38**	**87.21**	**52.31**	**44.04**	**8.30**	96.78	81.35	**67.19**	**27.25**	**89.65**	**63.48**	**74.47**	**39.16**

Success rates (%) are computed over 1024 rollouts. Both ResiP and KORR use the same base policy (DP) and are trained without disturbances. KORR consistently achieves higher task success and stronger robustness by incorporating simple Koopman dynamics.

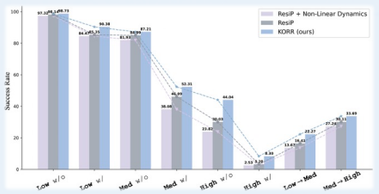
Methods	One_Leg		Round_Table		Lamp	
	→Med	↓Δ	→Med	↓Δ	→Med	↓Δ
ResiP [32]	16.41	83.28	7.32	92.39	17.28	80.04
KORR(ours)	**22.27**	77.44	**10.54**	89.11	**18.75**	79.09

Direct evaluation of Low-trained policies under Med initialization randomness.

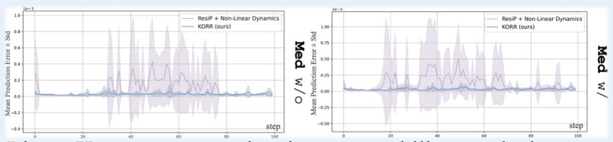


The performance gain remains consistent across different base policies.

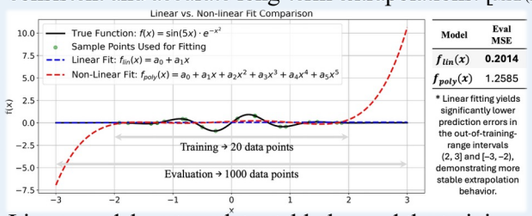
Linear Benefit Study



Nonlinear dynamics degrade performance, while Koopman-based dynamics consistently improve robustness and generalization.

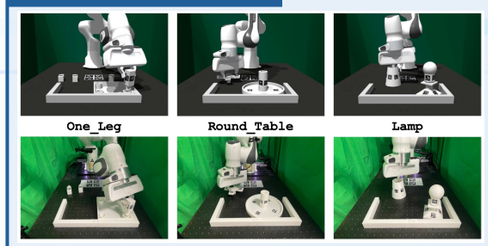


Linear Koopman constraints improve stability, producing more consistent and accurate long-term extrapolations. [MSE (mean ± std) /10]



Linear models extrapolate stably beyond the training range, while nonlinear models incur large errors.

Takeaways



- ☑ In certain settings with lifted latent spaces, Koopman-based linear representations can offer improved stability and robustness compared to nonlinear dynamics.
- ☑ Incorporating Koopman dynamics into residual reinforcement learning can improve performance, leading to enhanced robustness and partial generalization.
- ☑ Empirical results demonstrate consistent gains, validating the effectiveness of Koopman-guided refinement in robotic manipulation tasks.

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